

01

DEFINITION, PROPERTIES & IUB CLASSIFICATION

Enzymes are biocatalysts — proteins (except **ribozymes**, catalytic RNA) elaborated by living cells that **accelerate a reaction by lowering its activation energy**, without being consumed and **without altering ΔG or the equilibrium constant**. They only make equilibrium arrive sooner.

Protein nature — thermolabile, denatured by pH extremes, non-dialysable

Efficiency $10^4\text{--}10^{12}$ fold acceleration

Active in trace amounts

Highly specific

Reaction reversible, enzyme unchanged

Regulatable — allosteric, covalent, induction

EC CLASS	REACTION CATALYSED	SUB-GROUPS	VETERINARY EXAMPLE
1 · Oxidoreductases	Transfer of electrons / hydrogen (redox)	Dehydrogenases, oxidases, peroxidases, catalase, oxygenases	LDH, SDH, cytochrome oxidase, glutathione peroxidase
2 · Transferases	Transfer of a group (other than H) between molecules	Transaminases, kinases, methyl/acyl transferases	AST, ALT, hexokinase, GGT
3 · Hydrolases	Cleavage of a bond by addition of water	Esterases, glycosidases, peptidases, phosphatases	Lipase, amylase, trypsin, ALP, cholinesterase, urease
4 · Lyases	Removal of a group non-hydrolytically , leaving a double bond (or the reverse addition)	Decarboxylases, aldolases, hydratases, synthases	Aldolase, fumarase, carbonic anhydrase, HMG-CoA lyase
5 · Isomerases	Intramolecular rearrangement	Racemases, epimerases, mutases	Phosphohexose isomerase, phosphoglucomutase, methylmalonyl-CoA mutase
6 · Ligases (synthetases)	Joining of two molecules, coupled to ATP hydrolysis	C–C, C–N, C–O, C–S ligases	Pyruvate carboxylase, glutamine synthetase, DNA ligase, acetyl-CoA carboxylase

Six classes in order — "Over The Hill, Little Indian Lady": Oxidoreductase · Transferase · Hydrolase · Lyase · Isomerase · Ligase. EC number has four digits: class . sub-class . sub-sub-class . serial no. — e.g. EC 1.1.1.27 = lactate dehydrogenase.

Trap: a **synthase** (lyase, class 4) needs **no** ATP; a **synthetase** (ligase, class 6) **must** use ATP. Similarly **oxidase** uses O_2 as acceptor, **oxygenase** incorporates O_2 into the product, **dehydrogenase** uses NAD/FAD and never O_2 .

02

COFACTOR · COENZYME · PROSTHETIC GROUP

Apoenzyme
protein, inactive

+

Cofactor
metal or coenzyme

=

Holoenzyme
active

Coenzyme Loosely, **non-covalently bound**, dissociates freely, behaves as a **second substrate / co-substrate** and must be regenerated — NAD⁺, NADP⁺, CoA, THF.

Prosthetic group **Tightly or covalently bound**, stays with the enzyme through the cycle — FAD, biotin, haem, pyridoxal phosphate, lipoic acid.

Activator Metal ion required for activity or for substrate binding (metal-activated enzymes); if firmly held the enzyme is a **metalloenzyme**.

VITAMIN	COENZYME	REACTION TYPE	DEFICIENCY IN ANIMALS
B ₁ Thiamine	TPP	Oxidative decarboxylation, transketolase	Polioencephalomalacia (thiaminase, ruminants); Chastek paralysis — raw fish, fox/mink/cat
B ₂ Riboflavin	FMN, FAD	Hydrogen transfer	Curled-toe paralysis in chicks
B ₃ Niacin	NAD ⁺ , NADP ⁺	Hydride transfer	Black tongue in dogs; from tryptophan 60 : 1
B ₅ Pantothenate	CoA, ACP	Acyl transfer	Goose-stepping in pigs, dermatitis
B ₆ Pyridoxine	PLP	Transamination, decarboxylation, deamination, ALA synthase	Convulsions, microcytic anaemia
B ₇ Biotin	Biocytin	Carboxylation (CO ₂ fixation)	Bound by avidin of raw egg white; foot-pad lesions in poultry
B ₉ Folate	THF	One-carbon transfer	Megaloblastic anaemia; sulphonamide target in bacteria
B ₁₂ Cobalamin	Methyl- & deoxyadenosyl-cobalamin	Methylmalonyl-CoA mutase, homocysteine methylation	Cobalt deficiency in ruminants — blocks propionate → glucose; "pine/wasting disease"
Lipoic acid	Lipoamide	Acyl transfer in PDH / α -KGDH	—

Zn carbonic anhydrase, carboxypeptidase, LDH

Mg all kinases & ATP reactions

Mn arginase

Cu cytochrome oxidase, tyrosinase, SOD, ceruloplasmin

Fe catalase, peroxidase, cytochromes

Se glutathione peroxidase — **white muscle disease**

Mo xanthine oxidase

Ni urease

04

ENZYME-SUBSTRATE COMPLEX, THE MODERN ACTIVE CENTRE & SPECIFICITY

E + S

⇌

ES

→

EP

→

E + P

Two models of the active site:

Lock & key (Emil Fischer, 1894) Active site is a **rigid, pre-formed template** exactly complementary to the substrate. Explains **specificity** well, but not the action of enzymes on several related substrates.

Induced fit (Koshland, 1958) **The modern concept.** The active site is **flexible**; the approaching substrate **induces a conformational change** that brings the catalytic groups into correct alignment — like a hand entering a glove. Explains allosteric regulation, competitive inhibition and non-productive binding.

Features of the active centre: ① a small region — only **3–5 % of the total enzyme volume**; ② a **crevice or cleft** from which water is largely excluded; ③ built from amino-acid residues that are **far apart in the primary sequence but brought together by folding**; ④ has a **substrate-binding (contact) site** and a **catalytic site**; ⑤ binding is by weak, non-covalent forces — H-bond, ionic, hydrophobic, van der Waals; ⑥ residues commonly involved: **serine, histidine, cysteine, aspartate, glutamate, lysine, tyrosine** (the **Ser-His-Asp catalytic triad** of trypsin and chymotrypsin); ⑦ the shape complements the **transition state**, not the ground-state substrate.

TYPE OF SPECIFICITY	MEANING	EXAMPLE
Absolute	Acts on one substrate only	Urease — urea; carbonic anhydrase; lactase
Substrate (relative)	Acts on a group of closely related substrates	Alcohol dehydrogenase on several alcohols; hexokinase on glucose, fructose, mannose
Group	Acts on a particular chemical group next to the bond	Trypsin — carboxyl side of lysine & arginine ; chymotrypsin — aromatic residues; pepsin — aromatic
Bond	Acts on a bond type regardless of the rest	Esterases (ester), glycosidases (glycosidic), peptidases (peptide), lipases
Stereo / optical	Acts on only one optical or geometric isomer	L-amino acid oxidase vs D-amino acid oxidase; α - vs β -glycosidase; fumarase acts on trans-fumarate but not cis-maleate ; only D -sugars are used
Dual / co-substrate	Two substrates required	Transaminases (amino acid + keto acid); dehydrogenases (substrate + NAD ⁺)

Most-asked one-liner: "**Stereospecificity of fumarase**" — it hydrates fumarate (trans) but never maleate (cis). This one example answers both "optical specificity" and "why enzymes are more selective than chemical catalysts".

Trap: **zymogens / proenzymes** are catalytically inactive precursors activated by **limited proteolysis**: pepsinogen → pepsin (HCl, then autocatalysis), **trypsinogen** → **trypsin by enteropeptidase**, chymotrypsinogen, procarboxypeptidase, prothrombin, plasminogen. Premature activation inside the pancreas = **acute pancreatitis**.

85 FACTORS INFLUENCING ENZYME ACTION

① **Substrate concentration.** At low $[S]$ the rate is **first order** ($\propto [S]$); as $[S]$ rises the curve flattens to a **rectangular hyperbola** and becomes **zero order** at saturation, when every active site is occupied $\rightarrow V_{\max}$.

$$\text{Michaelis-Menten } v = V_{\max} [S] / (K_m + [S])$$

K_m = the substrate concentration at which $v = \frac{1}{2}V_{\max}$ (units of concentration, mol/L).

- Low K_m = high affinity for the substrate (hexokinase K_m 0.1 mM) and vice versa (glucokinase 10 mM).
- K_m is a **constant characteristic of the enzyme-substrate pair** and is **independent of enzyme concentration**; $V_{\max}$ is **directly proportional** to it.
- Lineweaver-Burk (double reciprocal) plot linearises the curve: $1/v = (K_m/V_{\max})(1/[S]) + 1/V_{\max}$ — y-intercept $1/V_{\max}$, x-intercept $-1/K_m$, slope $K_m/V_{\max}$.

② **Enzyme concentration.** With substrate in excess the velocity is **directly proportional to $[E]$** — a straight line through the origin. This linearity is the basis of every enzyme assay in the clinical laboratory.

③ **Temperature.** Rate roughly **doubles for each 10 °C rise ($Q_{10} \approx 2$)** up to the **optimum** ($\approx 37\text{--}40$ °C in mammals, $41\text{--}42$ °C in birds), beyond which thermal denaturation causes a sharp fall. The curve is therefore **bell-shaped, but asymmetrical**. Enzymes are inactivated above ~ 60 °C; hence samples are refrigerated.

④ **pH.** Bell-shaped curve; extremes alter the ionisation of the active site and of the substrate, and finally denature the enzyme.

Pepsin 1.5–2.0

Acid phosphatase 5

Salivary amylase 6.8

Trypsin / lipase 8

ALP 9–10

Arginase 9.5

⑤ **Others:** time of incubation, accumulation of product (product inhibition), activators (metal ions, bile salts for lipase), inhibitors, light and UV/ionising radiation.

Enzyme units — learn all four definitions verbatim.

International Unit (IU)

The amount of enzyme that converts **1 μmol of substrate per minute** under defined optimum conditions of temperature, pH and substrate concentration.

Katal (SI unit)

The amount of enzyme that converts **1 mole of substrate per second**. 1 katal = 6×10^7 IU; 1 IU = 16.67 nkat.

Specific activity

Units of enzyme activity per milligram of protein — the index of **purity**; it rises at every purification step.

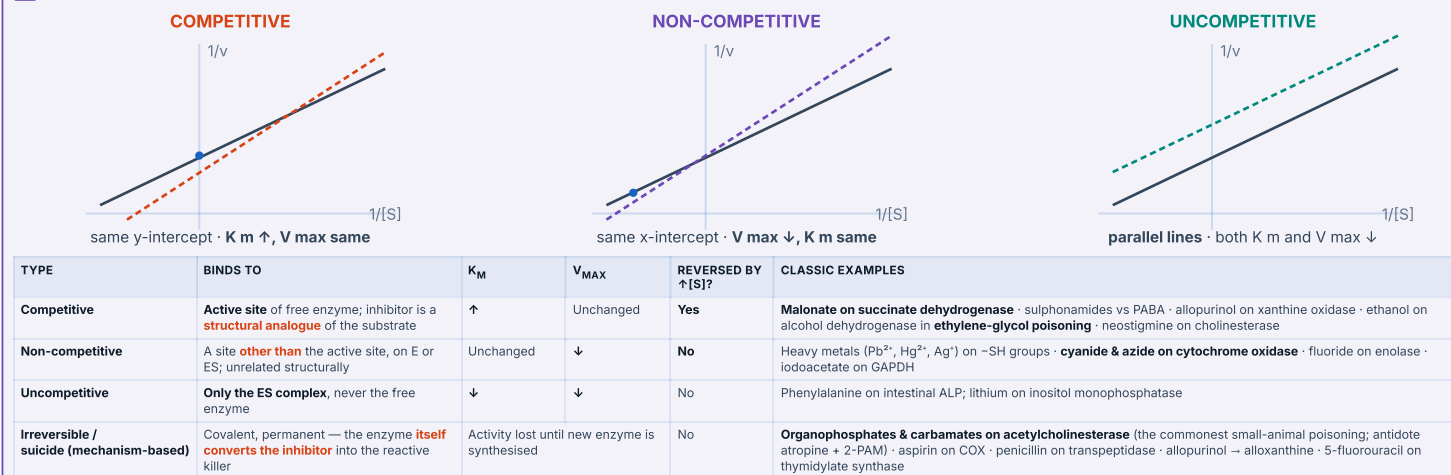
Turnover number

(k_{cat} , molecular activity)

The **number of substrate molecules converted per molecule of enzyme (or per active site) per unit time when the enzyme is fully saturated**. Highest known: **carbonic anhydrase $\approx 600\,000$ per second**.

Trap: specific activity measures **purity**; turnover number measures **catalytic efficiency** of a single molecule. Students routinely swap the two — a guaranteed one-mark loss.

96 ENZYME INHIBITION — THE FOUR TYPES, DECIDED BY THE LINEWEAVER-BURK PLOT



The two answers examiners want in full: ① **malonate vs succinate dehydrogenase** — malonate differs from succinate by one CH_2 group, occupies the active site, raises apparent K_m , and inhibition is relieved by excess succinate. ② **Organophosphate poisoning** — irreversible phosphorylation of the serine of acetylcholinesterase, "ageing" of the enzyme, and the rationale of oxime therapy **before** ageing occurs.

87 ALLOSTERIC ENZYMES & THE FOUR MODES OF METABOLIC REGULATION

Allosteric ("other site") enzymes are the regulatory enzymes of a pathway. They are **oligomeric**, possess a **catalytic site plus a separate allosteric site**, and bind their effectors non-covalently and reversibly.

- They give a **sigmoid (S-shaped) v vs $[S]$ curve, not a hyperbola** — because subunits show **co-operativity**; they therefore **do not obey Michaelis-Menten kinetics**.
- Positive effectors (activators)** shift the curve left ($\uparrow$ affinity); **negative effectors (inhibitors)** shift it right.
- Homotropic** = the substrate is itself the effector (O_2 on haemoglobin); **heterotropic** = a different molecule (ATP on PFK-1).
- Models: **MWC concerted / symmetry model** (all subunits flip $T \rightleftharpoons R$ together) and **KNF sequential model** (one at a time).
- Almost always the **first committed, irreversible, rate-limiting step** of the pathway, and usually subject to **feedback (end-product) inhibition**.

Learn these five allosteric enzymes with their effectors: **PFK-1** (– ATP, citrate; + AMP, **fructose-2,6-bisphosphate**) · **pyruvate carboxylase** (+ acetyl-CoA) · **glycogen phosphorylase** (+ AMP in muscle; – ATP, G6P) · **carbamoyl phosphate synthetase I** (+ N-acetylglutamate) · **HMG-CoA reductase** and **acetyl-CoA carboxylase** (+ citrate, – palmitoyl-CoA).

MODE OF REGULATION	SPEED	MECHANISM & EXAMPLE
Allosteric modulation	Seconds	Non-covalent binding of an effector; feedback inhibition. PFK-1, ATCase
Covalent modification	Seconds–minutes	Phosphorylation / dephosphorylation by hormone-driven kinases & phosphatases. Glycogen phosphorylase ON, glycogen synthase OFF when phosphorylated (glucagon/adrenaline); insulin does the reverse
Proteolytic (zymogen) activation	Minutes, irreversible	Digestive enzymes, clotting cascade, complement
Induction / repression	Hours–days	Change in the amount of enzyme by altered transcription. Glucokinase and lipogenic enzymes induced by insulin ; gluconeogenic enzymes by cortisol & glucagon ; cytochrome P450 induced by phenobarbitone
Compartmentation	—	Opposing pathways kept apart: β-oxidation in the mitochondrion, fatty-acid synthesis in the cytosol

Trap: phosphorylation does **not** mean "activation". It activates glycogen phosphorylase and hormone-sensitive lipase but **inactivates** glycogen synthase, acetyl-CoA carboxylase and pyruvate dehydrogenase. Remember it as "phosphorylation switches the body to the fasting mode".

08 ENZYMES & COENZYMES OF OXIDATION-REDUCTION

Biological oxidation = removal of electrons (usually as hydrogen atoms) from a substrate, coupled to reduction of an acceptor. It proceeds in **small steps down a redox-potential gradient** so the energy is captured, not lost as a flash of heat.

ENZYME CLASS	WHAT IT DOES	EXAMPLE
Dehydrogenases	Transfer H to NAD⁺/NADP⁺ or FAD/FMN , never to O ₂	LDH, MDH, isocitrate DH, succinate DH
Oxidases	Use O₂ as the direct acceptor → H ₂ O or H ₂ O ₂	Cytochrome oxidase (→H ₂ O); xanthine oxidase, D-amino acid oxidase (→H ₂ O ₂)
Peroxidases & catalase	Dispose of H ₂ O ₂ ; catalase: 2H ₂ O ₂ → 2H ₂ O + O ₂	Glutathione peroxidase (Se), myeloperoxidase, catalase (Fe)
Oxygenases	Dioxygenase — both O atoms into substrate; Monooxygenase / mixed-function oxidase — one O into substrate, one into water	Cytochrome P450 (steroidogenesis, drug metabolism); homogentisate dioxygenase

NAD⁺ / NADP⁺ From **niacin**; accept a **hydride ion (2e⁻ + H⁺)**. NADH → ETC for ATP; NADPH → **reductive biosynthesis** and antioxidant defence.

FAD / FMN From **riboflavin**; accept **2H (2e⁻ + 2H⁺)**; tightly bound prosthetic groups.

Coenzyme Q (ubiquinone) **Lipid-soluble, freely mobile** in the membrane; the **collection point** for electrons from Complex I, Complex II, ETF of β-oxidation and glycerol-3-phosphate dehydrogenase. Not protein, not vitamin-derived.

Cytochromes Haemoproteins; transfer **one electron at a time** by Fe³⁺ ⇌ Fe²⁺. Order b → c₁ → c → a → a₃. **Cytochrome c is the only water-soluble, mobile one.**

Iron-sulphur proteins Non-haem iron (Fe-S centres) in Complexes I, II, III; one-electron carriers.

-7.3
ATP, kcal/mol

-14.8
PEP — highest

-11.8
1,3-BPG

-10.3
creatine-P

09 THE RESPIRATORY CHAIN — FOUR COMPLEXES

INTERMEMBRANE SPACE — H⁺ accumulates (low pH, positive)

MATRIX

NADH → FMN → CoQ → cyt b → c₁ → c → a → a₃ → ½O₂ + 2H⁺ → H₂O

Rotenone/amytal ⊥ I · Malonate/carboxin ⊥ II · Antimycin A/BAL ⊥ III
Cyanide, CO, azide, H₂S ⊥ IV · Oligomycin ⊥ V · Attractyloside ⊥ ADP/ATP translocase

Electrons always flow from the **more negative to the more positive redox potential**;
Complex II pumps **no** protons — that is why FADH₂ yields less ATP than NADH.

10 OXIDATIVE PHOSPHORYLATION, UNCOUPLERS & THE SHUTTLES

Chemiosmotic theory — Peter Mitchell, 1961 (Nobel 1978). Electron flow through Complexes I, III and IV pumps **H⁺ from the matrix into the intermembrane space**, creating a **proton-motive force** made of a pH gradient (ΔpH) plus a membrane potential (Δψ). Protons re-enter **only** through the F₀ channel of **ATP synthase (Complex V)**, and this flow drives **rotational catalysis** in F₁ (Boyer's binding-change mechanism) which releases ATP.

- **Prerequisite: an intact, impermeable inner mitochondrial membrane.** Any leak abolishes ATP synthesis while electron flow continues.
- **Respiratory control:** the rate of the chain is governed by the availability of **ADP** — the cell makes ATP only as fast as it spends it.
- **P : O ratio** = molecules of ATP formed per atom of oxygen reduced: **NADH = 2.5 (classically 3), FADH₂ = 1.5 (classically 2).**

Uncouplers carry H⁺ back across the membrane, short-circuiting the gradient. **Electron transport continues (even accelerates) but ATP is not made — the energy appears as heat.** Signs: hyperthermia, tachypnoea, weight loss, no ATP gain.

2,4-Dinitrophenol, dinitro-o-cresol

CCCP, FCCP

Salicylate in overdose

Unconjugated bilirubin

Excess thyroxine, free fatty acids

Thermogenin (UCP-1) — physiological

Veterinary gold: brown adipose tissue of the newborn lamb, piglet, kid and kitten is packed with **UCP-1/thermogenin**, a **deliberate, hormone-controlled uncoupler** that generates heat instead of ATP — **non-shivering thermogenesis**. Its failure or absence explains neonatal hypothermia and much of the "weak-neonate" complex.

Trap: distinguish sharply — an **inhibitor** stops electron flow **and** ATP synthesis and oxygen consumption falls; an **uncoupler** lets electron flow and O₂ consumption **increase** while ATP falls; **oligomycin** blocks the ATP synthase itself, so both stop — and adding an uncoupler **restores** respiration after oligomycin. That last sentence is a favourite viva question.

Shuttles. The inner membrane is impermeable to NADH, so cytosolic NADH from glycolysis must be transferred indirectly — and the choice of shuttle decides the final ATP count for glucose.

SHUTTLE	DELIVERS AS	ATP	TISSUE
Glycerol-3-phosphate	FADH ₂ (via CoQ)	1.5	Skeletal muscle, brain, insect flight muscle
Malate-aspartate	NADH	2.5	Liver, heart, kidney — more efficient

32
ATP/glucose, malate-aspp

30
ATP/glucose, glycerol-P

38 / 36
classical figures

Other transporters of the inner membrane: adenine-nucleotide translocase (ADP in / ATP out — blocked by **atractyloside**), phosphate translocase, tricarboxylate (citrate) carrier, and the **carnitine shuttle** for fatty acids.

11

GLYCOLYSIS (EMBDEN-MEYERHOF PATHWAY) — CYTOSOL, EVERY CELL, AEROBIC OR ANAEROBIC

Glucose

hexokinase / glucokinase

→ -1 ATP →

G-6-P

→

F-6-P

PFK-1

→ -1 ATP RATE-LIMITING →

F-1,6-BP

aldolase →

2 × G-3-P

2 × G-3-P

GAPDH

+2 NADH →

2 × 1,3-BPG

+2 ATP (SLP) →

3-PG → 2-PG

enolase →

2 × PEP

pyruvate kinase

+2 ATP (SLP) →

2 × Pyruvate

- Three irreversible, regulatory steps: hexokinase/glucokinase, PFK-1 (the key rate-limiting enzyme), pyruvate kinase. These are exactly the three bypassed in gluconeogenesis.
- Balance sheet: 2 ATP invested, 4 ATP generated by substrate-level phosphorylation → net 2 ATP + 2 NADH per glucose. Anaerobically the 2 NADH are spent reducing pyruvate to lactate (LDH) to regenerate NAD⁺, so the net stays 2 ATP — the only ATP source for the RBC (no mitochondria), working muscle, cornea, renal medulla and rapidly growing tissue.
- Hexokinase vs glucokinase: hexokinase is in all tissues, K_m 0.1 mM, inhibited by G-6-P; glucokinase is in liver and β-cell, K_m ≈ 10 mM, not inhibited by G-6-P and induced by insulin — it only works after a meal, letting the liver take up glucose only when blood glucose is high.
- Rapoport-Luebering shunt in the RBC diverts 1,3-BPG to 2,3-BPG, which lowers haemoglobin's O₂ affinity and improves tissue delivery — at the cost of one ATP.

Practical link you will be asked: sodium fluoride inhibits enolase, so NaF-containing (grey-top) tubes are used for blood glucose collection — without it, erythrocyte glycolysis destroys ~10 % of the glucose per hour and gives a falsely low result. Also: iodoacetate blocks GAPDH, arsenate uncouples at GAPDH (glycolysis runs but makes no ATP).

Link reaction — pyruvate dehydrogenase complex (mitochondrial matrix): pyruvate + CoA + NAD⁺ → acetyl-CoA + CO₂ + NADH. A multienzyme complex of 3 enzymes needing 5 coenzymes: TPP, Lipoamide, CoA, FAD, NAD⁺. Irreversible — this single fact is why fatty acids can never be converted to glucose in animals. Inhibited by arsenite and by mercury (bind lipoamide -SH); switched off by phosphorylation (PDH kinase, activated by ATP/NADH/acetyl-CoA) and on by dephosphorylation (PDH phosphatase, Ca²⁺, insulin).

PDH coenzymes — "Tender Loving Care For Nancy": TPP, Lipoamide, CoA, FAD, NAD⁺. The α-ketoglutarate dehydrogenase complex uses exactly the same five.

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KREBS / TCA / CITRIC ACID CYCLE

Site: mitochondrial matrix (succinate dehydrogenase alone is in the inner membrane). Nature: the final common oxidative pathway for carbohydrate, fat and protein, and amphibolic — it both degrades and supplies precursors. Strictly aerobic (needs NAD⁺/FAD regenerated by the ETC), though O₂ takes no direct part.

STEP	ENZYME	YIELD / NOTE
Acetyl-CoA + OAA	Citrate synthase	Condensation; the entry point
Citrate → isocitrate	Aconitase	Blocked by fluoroacetate ("1080", Dichapetalum/Gifblaar plant poisoning)
Isocitrate → α-KG	Isocitrate DH — rate limiting	1st NADH + 1st CO ₂
α-KG → succinyl-CoA	α-KG dehydrogenase complex	2nd NADH + 2nd CO ₂ ; same 5 coenzymes as PDH
Succinyl-CoA → succinate	Succinyl-CoA synthetase	GTP by substrate-level phosphorylation — the only one in the cycle
Succinate → fumarate	Succinate DH (Complex II)	FADH ₂ ; membrane bound; inhibited by malonate
Fumarate → malate	Fumarase	Stereospecific hydration
Malate → OAA	Malate DH	3rd NADH; OAA regenerated

3
NADH

1
FADH₂

1
GTP

10
ATP per acetyl-CoA

2
CO₂ released

- Anaplerotic ("topping-up") reactions replace intermediates drained for biosynthesis: pyruvate carboxylase (pyruvate → OAA, biotin, activated by acetyl-CoA) — the chief one; transamination of glutamate/aspartate; and in the ruminant, propionate → succinyl-CoA.
- Regulation: by the ATP/ADP and NADH/NAD⁺ ratios at citrate synthase, isocitrate DH and α-KG DH; Ca²⁺ activates all three during muscular work.
- Amphibolic outputs: citrate → fatty acid synthesis; α-KG & OAA → amino acids & gluconeogenesis; succinyl-CoA → haem synthesis; fumarate ← urea cycle.

13

HMP SHUNT (PENTOSE PHOSPHATE PATHWAY)

Site: cytosol of liver, adipose tissue, lactating mammary gland, adrenal cortex, gonads, RBC and leucocytes — all tissues that need NADPH. Also called the Warburg-Dickens-Horecker or hexose monophosphate pathway. It produces no ATP and consumes none.

G-6-P

G-6-P dehydrogenase

· rate-limiting · NADPH #1 →

6-phosphogluconate

6-PG DH · NADPH #2 + CO₂ →

Ribulose-5-P

Ribulose-5-P

→

Ribose-5-P

non-oxidative, reversible: transketolase (TPP) & transaldolase →

F-6-P + G-3-P back to glycolysis

- Two products, both irreplaceable: NADPH (2 per G-6-P; 12 per glucose fully cycled) and ribose-5-phosphate for nucleotides, ATP, NAD, FAD, CoA.
- What NADPH is spent on: ① fatty acid, cholesterol and steroid-hormone synthesis — hence the pathway's intense activity in adipose tissue, mammary gland and adrenal cortex; ② reduction of glutathione (glutathione reductase) → GSH, the RBC's shield against oxidant damage; ③ the respiratory burst of neutrophils via NADPH oxidase; ④ cytochrome P450 hydroxylations in detoxification; ⑤ reduction of folate and of methaemoglobin.

Clinical anchor: when the NADPH-glutathione shield is overwhelmed, haemoglobin is oxidised → Heinz-body haemolytic anaemia and methaemoglobinemia. Classic Indian-practice causes: onion and Brassica/kale poisoning in cattle and dogs, paracetamol in the cat, red maple in the horse, and phenothiazine.

14

GLUCONEOGENESIS · CORI CYCLE · GLYCOGEN METABOLISM · BIOENERGETICS

Gluconeogenesis — synthesis of glucose from non-carbohydrate sources. Site: liver (≈90 %) and renal cortex (≈10 %); spans cytosol, mitochondrion and endoplasmic reticulum. Substrates: lactate, glycerol, propionate (the ruminant's chief one), and glucogenic amino acids — alanine above all.

BYPASS ENZYME	REACTION & NOTE
Pyruvate carboxylase	Pyruvate → OAA; mitochondrial, biotin, needs acetyl-CoA as obligatory activator
PEP carboxykinase	OAA → PEP; uses GTP; induced by cortisol & glucagon
Fructose-1,6-bisphosphatase	Rate-limiting; inhibited by AMP and by fructose-2,6-bisphosphate
Glucose-6-phosphatase	In the ER of liver and kidney only — muscle lacks it, which is why muscle glycogen can never raise blood glucose

Cost: 6 high-energy phosphates (4 ATP + 2 GTP) plus 2 NADH per glucose — gluconeogenesis is expensive, which is why the liver will not run it and glycolysis at the same time (reciprocal regulation).

Golden rule: acetyl-CoA (and therefore even-chain fatty acids) can never be converted to glucose — the PDH step is irreversible. Only glycerol and odd-chain/propionate carbons from fat are glucogenic.

Cori cycle (lactic acid cycle). Muscle glycolysis under anaerobic work → lactate → blood → liver → gluconeogenesis → glucose → back to muscle. It shifts the metabolic burden of the muscle to the liver and clears lactic acid, at a net cost of 4 ATP to the liver for every 2 recovered by muscle.

Muscle glucose → lactate

→

blood

→

Liver lactate → glucose

→

blood → muscle

The parallel glucose-alanine (Cahill) cycle carries both the carbon and the amino nitrogen of muscle protein to the liver, where alanine is transaminated to pyruvate for gluconeogenesis and its nitrogen enters the urea cycle.

Glycogenesis. Glucose → G-6-P → G-1-P (phosphoglucomutase) → UDP-glucose (uses UTP) → glycogen. Glycogen synthase makes the α1→4 chain on the primer protein glycogenin; branching enzyme (amylo-4,6-transglucosidase) transfers blocks of ~7 residues to make α1→6 branches. Activated by insulin (dephosphorylation).

Glycogenolysis. Glycogen phosphorylase (rate-limiting, needs PLP) carries out phosphorolysis, releasing G-1-P; the debranching enzyme (transferase + α1,6-glucosidase) releases the ~8 % of residues at branch points as free glucose. Driven by glucagon and adrenaline via cAMP → protein kinase A, which simultaneously switches synthase off — one signal, opposite effects, no futile cycle.

Ruminant special (worth a whole long answer): almost no glucose is absorbed from the gut, so the ruminant runs gluconeogenesis continuously, all day, mainly from propionate: propionate → propionyl-CoA → methylmalonyl-CoA →(B₁₂, mutase)→ succinyl-CoA → OAA → glucose. Cobalt/vitamin B₁₂ deficiency blocks this step and causes wasting; and any surge in demand (peak lactation, twin pregnancy) that outruns propionate supply produces ketosis and pregnancy toxemia — the whole of Unit 3's energy section rests on this paragraph.

PROCESS	ATP
Glycolysis, anaerobic	+2
Glycolysis, aerobic (to pyruvate)	+5 or +7
PDH ×2	+5
TCA ×2 acetyl-CoA	+20
Glucose → CO ₂ + H ₂ O	32 (30 via glycerol-P shuttle)
Gluconeogenesis, cost	-6
Glycogen → lactate (from G-1-P)	+3

Trap: glycogen storage diseases — von Gierke (type I, glucose-6-phosphatase; severe fasting hypoglycaemia + hepatomegaly), Pompe (type II, lysosomal α-glucosidase; reported in cattle and cats), McArdle (type V, muscle phosphorylase; exercise intolerance with no rise in blood lactate).

15

B-OXIDATION OF FATTY ACIDS (KNOOP)

Fatty acid
cytosol

acyl-CoA synthetase (thiokinase) · ATP→AMP+PPI = 2 ~P spent →

Acyl-CoA

carnitine shuttle: CPT-I (rate-limiting, inhibited by malonyl-CoA) → translocase → CPT-II → Matrix

Four repeating steps, each cycle shortening the chain by two carbons — mnemonic "Ox-Hy-Ox-Th":

1. **Oxidation** — acyl-CoA dehydrogenase → FADH₂
2. **Hydration** — enoyl-CoA hydratase
3. **Oxidation** — β-hydroxyacyl-CoA dehydrogenase → NADH
4. **Thiolysis** — thiolase → **acetyl-CoA** + acyl-CoA (n-2)

Palmitate (C16) bioenergetics — memorise the arithmetic: 7 cycles → 8 acetyl-CoA + 7 FADH₂ + 7 NADH. (8 × 10) + (7 × 1.5) + (7 × 2.5) = 80 + 10.5 + 17.5 = 108 - 2 (activation) = 106 ATP (classical figures: 129 - 2 = 129). Cycles = (n/2) - 1; acetyl-CoA = n/2.

- **Carnitine** is synthesised from **lysine + methionine** and is the obligatory ferry for long-chain fatty acids; short and medium chains enter freely. **Malonyl-CoA inhibits CPT-I** — the elegant switch that stops the cell burning and building fat at once.
- **Odd-chain fatty acids** end in **propionyl-CoA** → **methylmalonyl-CoA** → (B₁₂) → **succinyl-CoA** — the only part of a fatty acid that is **glucogenic**. Unsaturated fatty acids need an extra **isomerase** and **reductase**. **Very-long-chain fatty acids** are shortened in **peroxisomes**; α-oxidation handles branched phytanic acid; ω-oxidation is a minor microsomal route.

16

KETONE-BODY FORMATION & FATTY ACID BIOSYNTHESIS

Ketogenesis — liver mitochondria only, by the HMG-CoA pathway.

2 Acetyl-CoA

→

Acetoacetyl-CoA

HMG-CoA synthase — rate-limiting →

HMG-CoA

HMG-CoA lyase →

Acetoacetate

Acetoacetate

BHB dehydrogenase, NADH →

β-hydroxybutyrate — the major one

· spontaneous decarboxylation →

Acetone — exhaled, "sweet breath"

- **The liver makes ketones but cannot use them** — it lacks **thiophorase (succinyl-CoA : acetoacetate CoA transferase)**. Heart, skeletal muscle, renal cortex and, in prolonged starvation, the **brain** oxidise them; ketone bodies are water-soluble, need no carrier and **spare glucose**.
- **Trigger in every species:** low insulin / high glucagon → unrestrained lipolysis → acetyl-CoA floods the liver, while **oxaloacetate is drained away into gluconeogenesis**, so acetyl-CoA cannot enter the TCA cycle. Hence the classical statement — "**fat burns in the flame of carbohydrate**."
- **Causes:** starvation, high-fat/low-carbohydrate diet, **diabetes mellitus**, **bovine ketosis** of early lactation, **pregnancy toxemia** of the ewe and doe.

Ruminant twist: β-hydroxybutyrate is a **normal** ruminant metabolite — the rumen epithelium converts absorbed butyrate to BHB during every meal. Ketosis is therefore a matter of degree, and BHB is the field test: **<1.0 mmol/L normal · 1.2-1.4 subclinical · >2.6-3.0 clinical ketosis**.

Fatty acid biosynthesis (de novo lipogenesis) — **cytosol**. Acetyl-CoA is exported from the mitochondrion as citrate (ATP-citrate lyase). **Acetyl-CoA carboxylase is the rate-limiting enzyme**: it needs **biotin, CO₂ and ATP**, is activated by **citrate** and **insulin** and inhibited by **palmitoyl-CoA, glucagon and adrenaline**, and it makes **malonyl-CoA**. **Fatty acid synthase**, a multienzyme dimer carrying **ACP (phosphopantetheine, from pantothenic acid)**, then repeats condensation → reduction → dehydration → reduction and **stops at palmitate (C16)**. Per palmitate: **8 acetyl-CoA, 7 ATP, 14 NADPH** (from the HMP shunt and malic enzyme). Elongation and desaturation (Δ9, Δ6, Δ5 only) follow in the ER.

Species trap: in ruminants the principal site of lipogenesis is adipose tissue and the mammary gland, not the liver, and the starting material is **acetate absorbed from the rumen**, not citrate-derived acetyl-CoA. This is why a low-fibre ration (less acetate) causes **milk-fat depression**.

17

PROTEIN METABOLISM — BIOSYNTHESIS, DEGRADATION AND THE FATE OF THE AMINO GROUP

Amino acid pool & nitrogen balance. Dietary protein, tissue protein turnover and de-novo synthesis feed a common pool (≈100 g in an adult) drawn on for protein synthesis, hormones, purines, haem, creatine, and for energy.

Positive balance = growth, pregnancy, lactation, convalescence; **negative balance** = starvation, fever, trauma, parasitism, protein-deficient ration.

Degradation follows two routes: the **ATP-dependent ubiquitin-proteasome system** in the cytosol for short-lived and abnormal proteins (tagged by polyubiquitin), and **lysosomal cathepsins** for extracellular and membrane proteins. **Biosynthesis** is ribosomal translation of mRNA (see Sheet 6).

Transamination Reversible transfer of an α-amino group to an α-keto acid. Coenzyme **pyridoxal phosphate (B₆)**. **ALT**: alanine + α-KG ⇌ pyruvate + glutamate. **AST**: aspartate + α-KG ⇌ OAA + glutamate. **Every amino acid except lysine, threonine, proline and hydroxyproline** undergoes it. Its purpose is to **funnel all nitrogen into glutamate**. Leakage of ALT/AST into serum is the basis of liver and muscle profiling in Unit 3.

Deamination **Oxidative** — **glutamate dehydrogenase** (mitochondrial, NAD(P)⁺, reversible, allosterically inhibited by ATP/GTP and activated by ADP/GDP) is the chief route and the only one that liberates ammonia rapidly; L- and D-amino acid oxidases (FMN/FAD) are minor and give H₂O₂. **Non-oxidative** — serine/threonine dehydratase, cysteine desulphydrase, histidase.

Transdeamination **Transamination + oxidative deamination coupled** — the combined route by which the nitrogen of almost all amino acids finally reaches ammonia.

Decarboxylation (PLP-dependent) removes —COOH as CO₂ and yields the **biogenic amines** — small marks, frequently asked:

AMINO ACID	AMINE	ACTION
Histidine	Histamine	Allergy, gastric HCl, vasodilation
Tryptophan	Serotonin → melatonin	Neurotransmitter, gut motility
Tyrosine	Dopamine → NA → adrenaline; DOPA → melanin	Catecholamines, pigment
Glutamate	GABA	Chief inhibitory transmitter
Cysteine	Taurine	Bile salts; essential in the cat
Ornithine / lysine	Putrescine, spermine, spermidine, cadaverine	Polyamines — cell proliferation
Serine	Ethanolamine → choline	Phospholipid, acetylcholine

Amines are inactivated by **monoamine and diamine oxidases**. In the gut, bacterial decarboxylation of unabsorbed amino acids yields putrefactive amines — relevant to Unit 3 digestive disorders.

Ammonia transport. Free NH₃ is intensely neurotoxic (it drains α-ketoglutarate from the brain's TCA cycle and swells astrocytes as glutamine accumulates). It travels in blood in two safe forms: **glutamine** (glutamine synthetase, ATP; released by glutaminase in liver and kidney) — the principal carrier — and **alanine** from muscle by the glucose-alanine cycle.

Urea cycle (Krebs-Henseleit, 1932) — liver only; steps 1-2 mitochondrial, 3-5 cytosolic.

#	ENZYME	PRODUCT / NOTE
1	Carbamoyl phosphate synthetase I	NH ₃ + CO ₂ + 2 ATP: rate-limiting ; absolutely requires N-acetylglutamate as activator
2	Ornithine transcarbamoylase	+ ornithine → citrulline
3	Argininosuccinate synthetase	+ aspartate (the 2nd nitrogen); ATP → AMP + PPI = 2 ~P
4	Argininosuccinate lyase	→ arginine + fumarate — the link to the TCA cycle ("Krebs bicycle")
5	Arginase (Mn ²⁺ , liver-specific)	→ UREA + ornithine (recycled)

Remember: Ordinarily, Careless Crappers Are Also Frivolous About Urination — Ornithine, Carbamoyl-P, Citrulline, Argininosuccinate, Arginine, Fumarate, Urea. Cost: **4 ATP equivalents (3 ATP molecules) per urea**. One nitrogen from free ammonia, one from **aspartate**; the carbon from CO₂.

Clinical payoffs: ① **Portosystemic shunt** in the dog — high blood ammonia, low BUN, ammonium biurate crystals, hepatic encephalopathy. ② **Urea poisoning in ruminants** — rumen urease liberates NH₃ faster than the liver can clear it; rumen pH rises, absorption accelerates; treat with **cold water + 5 % acetic acid (vinegar)**. ③ **Nitrogen recycling** — urea returns to the rumen in saliva and across the wall, where microbes reuse it, so ruminants thrive on non-protein nitrogen. ④ Ureotelic (mammals) vs **uricotelic (birds, reptiles)** vs ammonotelic (fish).

18 PURINE & PYRIMIDINE METABOLISM

Purine — de novo

Built **on the ribose**: R-5-P →(PRPP synthetase)—→ **PRPP** →(glutamine-PRPP amidotransferase — **committed, rate-limiting step, feedback-inhibited by AMP, GMP and IMP**)→ 11 steps → **IMP** → AMP (needs **GTP** + aspartate) or GMP (needs **ATP** + glutamine). The reciprocal ATP/GTP requirement keeps the two pools balanced. Atoms come from glycine, glutamine, aspartate, CO₂ and formyl-THF.

Purine — salvage

Purine — catabolism

Pyrimidine

The **ring is made first and attached to PRPP afterwards**. **CPS-II** (cytosolic, uses glutamine, **needs no N-acetylglutamate**) is rate-limiting → carbamoyl aspartate → dihydroorotate → orotate → OMP → **UMP** → **UTP** → **CTP**. dTMP comes from dUMP by **thymidylate synthase** using N⁵,N¹⁰-methylene-THF.

Deoxy-ribonucleotides

Ribonucleotide reductase reduces NDP → dNDP using **thioredoxin**; allosterically regulated; inhibited by hydroxyurea.

The three points that always appear: ① **Birds and reptiles are uricotelic** — they lack uricase and excrete uric acid as a paste; overload gives **avian visceral and articular gout** (high-protein ration, renal damage, water deprivation). ② The **Dalmatian dog** handles urate abnormally and forms **urate uroliths**. ③ **Allopurinol** is a **suicide inhibitor of xanthine oxidase**; **5-fluorouracil** blocks thymidylate synthase and **methotrexate** blocks dihydrofolate reductase — the classical antimetabolites. Pyrimidine catabolism, by contrast, gives soluble β-alanine and β-aminoisobutyrate, so **pyrimidines never form stones**.

28 REGULATION & INTEGRATION OF METABOLISM

Three junction molecules decide the fate of everything: **glucose-6-phosphate** (glycogen / glycolysis / HMP shunt), **pyruvate** (lactate / acetyl-CoA / OAA / alanine) and **acetyl-CoA** — the true crossroads, from which carbon can go to **TCA, fatty acids, cholesterol or ketone bodies but never back to glucose**.

ORGAN	METABOLIC CHARACTER
Liver	The glucostat and central processing plant — glycogen, gluconeogenesis, ketogenesis, urea, lipoproteins, detoxification
Brain	Obligate glucose user; switches to ketone bodies only after days of starvation; no fuel stores
Skeletal muscle	Glycogen for its own use only ; the body's protein reserve ; exports alanine and lactate
Adipose tissue	Triacylglycerol store; hormone-sensitive lipase is the tap
RBC	Anaerobic glycolysis only — no mitochondria
Kidney	Gluconeogenesis and ammoniogenesis in acidosis

Fed state (insulin dominant): glucose uptake via GLUT-4, **glycogenesis, lipogenesis, protein synthesis**; glycolysis and the HMP shunt run; PDH is dephosphorylated and active.

Fasting and starvation (glucagon, adrenaline, cortisol, GH): a strict sequence — ① **hepatic glycogenolysis** (first 12–24 h) → ② **gluconeogenesis** from amino acids, lactate and glycerol → ③ **lipolysis and ketogenesis**, which spare glucose for the brain → ④ **protein catabolism** as the last resort. Muscle switches to fatty acids and ketones within days.

HORMONE	EFFECT ON BLOOD GLUCOSE
Insulin	The only hypoglycaemic hormone — uptake, glycogenesis, lipogenesis
Glucagon	↑ glycogenolysis, gluconeogenesis, ketogenesis (liver only)
Adrenaline	↑ glycogenolysis in liver and muscle; lipolysis
Cortisol	↑ gluconeogenesis, protein catabolism, insulin resistance
GH, thyroxine, ACTH	Hyperglycaemic, anti-insulin

Ruminant integration — write this as your conclusion in any long answer: the ruminant absorbs virtually no glucose. Instead it absorbs **VFAs: acetate (≈65 %) → lipogenesis and milk fat; propionate (≈20 %) → the sole major gluconeogenic precursor; butyrate (≈15 %) → converted to β-hydroxybutyrate by the rumen wall**. Therefore the ruminant is permanently gluconeogenic, permanently mildly ketogenic, and any negative energy balance — **peak lactation, twin pregnancy, transport, anorexia** — tips it straight into **ketosis or pregnancy toxemia**. Everything in Unit 3's energy-disorder section is this paragraph applied clinically.

Trap: the ruminant's low blood glucose (45–60 mg/dL) is **normal**, not hypoglycaemia; and a "normal" glucose in a ketotic cow does not exclude ketosis — **BHB, not glucose, is the diagnostic analyte**.

21 FINAL RECALL BANK — READ THESE LAST, STRAIGHT BEFORE THE PAPER

Enzyme class of transaminases	Transferases (class 2)	Rate-limiting enzyme of glycolysis	Phosphofructokinase-1
Enzyme class needing ATP to join	Ligases / synthetases (6)	Glycolysis inhibitor used in glucose tubes	Sodium fluoride (enolase)
Apoenzyme + cofactor	Holoenzyme	Liver glucose sensor enzyme	Glucokinase (high K _m)
Coenzyme of transamination	Pyridoxal phosphate (B ₆)	Net ATP of anaerobic glycolysis	2
Coenzyme of carboxylation	Biotin	Coenzymes of the PDH complex	TPP, lipoamide, CoA, FAD, NAD ⁺
Vitamin bound by avidin	Biotin	Rate-limiting enzyme of TCA	Isocitrate dehydrogenase
Coenzyme of methylmalonyl-CoA mutase	Vitamin B ₁₂ (cobalt)	Only substrate-level phosphorylation in TCA	Succinyl-CoA synthetase (GTP)
Selenium enzyme	Glutathione peroxidase	Fluoroacetate poisoning blocks	Aconitase
Molybdenum enzyme	Xanthine oxidase	ATP per acetyl-CoA in TCA	10
LDH isoenzyme of heart / liver	LDH-1 (H ₄) / LDH-5 (M ₄)	Total ATP per glucose	32 (30 via glycerol-P shuttle)
ALP isoenzyme unique to dogs	Corticosteroid-induced C-ALP	Rate-limiting enzyme of HMP shunt	Glucose-6-P dehydrogenase
Modern concept of the active site	Koshland's induced fit	Two products of the HMP shunt	NADPH + ribose-5-phosphate
Enzyme with absolute specificity	Urease	Enzyme absent in muscle	Glucose-6-phosphatase
Enzyme showing stereospecificity	Fumarase (trans only)	Chief gluconeogenic substrate in ruminants	Propionate
Km equals	[S] at ½ V _{max}	ATP cost of gluconeogenesis	6 per glucose
Low Km means	High substrate affinity	Muscle lactate → liver glucose	Cori cycle
Km independent of	Enzyme concentration	Glycogen synthesis donor	UDP-glucose
Lineweaver-Burk x-intercept	-1/K _m	Glycogen primer protein	Glycogenin
Definition of 1 katal	1 mol substrate per second	Coenzyme of glycogen phosphorylase	Pyridoxal phosphate
1 katal in IU	6 × 10 ⁷ IU	Rate-limiting step of β-oxidation entry	Carnitine palmitoyl transferase-I
Index of enzyme purity	Specific activity	CPT-I is inhibited by	Malonyl-CoA
Highest turnover number	Carbonic anhydrase	ATP from complete oxidation of palmitate	106
Malonate inhibits	Succinate DH — competitive	Rate-limiting enzyme of ketogenesis	HMG-CoA synthase
Cyanide inhibits	Cytochrome oxidase (Complex IV)	Major ketone body	β-hydroxybutyrate
Organophosphates inhibit	Acetylcholinesterase — irreversible	Enzyme the liver lacks for ketone use	Thiophorase
Inhibition with parallel LB lines	Uncompetitive	Rate-limiting enzyme of lipogenesis	Acetyl-CoA carboxylase (biotin)
Kinetics of allosteric enzymes	Sigmoid, not Michaelis-Menten	Fatty acid synthase stops at	Palmitate, C16
Chemiosmotic theory given by	Peter Mitchell, 1961	NADPH needed per palmitate	14
Complex that pumps no protons	Complex II (SDH)	Chief oxidative deamination enzyme	Glutamate dehydrogenase
Inhibitor of ATP synthase	Oligomycin	Chief transporter of ammonia	Glutamine
Classical uncoupler	2,4-Dinitrophenol	Rate-limiting enzyme of urea cycle	CPS-I (needs N-acetylglutamate)
Physiological uncoupler	Thermogenin (UCP-1), brown fat	ATP spent per molecule of urea	4 high-energy equivalents
P : O ratio of NADH / FADH ₂	2.5 / 1.5	Liver-specific urea cycle enzyme	Arginase (Mn ²⁺)
		Committed step of purine synthesis	PRPP amidotransferase
		Enzyme inhibited by allopurinol	Xanthine oxidase
		Operon model proposed by	Jacob & Monod

Thirty-second closing sweep: OTHLIL six classes · Km = [S] at ½Vmax; competitive raises Km, non-competitive lowers Vmax · 1 katal = 6×10⁷ IU · malonate ⊥ SDH, cyanide ⊥ complex IV, oligomycin ⊥ complex V, DNP uncouples · PFK-1 in, F-1,6-BPase out · isocitrate DH is TCA's rate-limiter; 10 ATP per acetyl-CoA · G-6-PD makes NADPH; onion & paracetamol break the shield · CPT-I blocked by malonyl-CoA; palmitate = 106 ATP · HMG-CoA synthase makes ketones, liver lacks thiophorase · CPS-I needs N-acetylglutamate; 4 ~P per urea · acetyl-CoA can never become glucose · propionate is the ruminant's glucose.